

Dropout

Deep Learning

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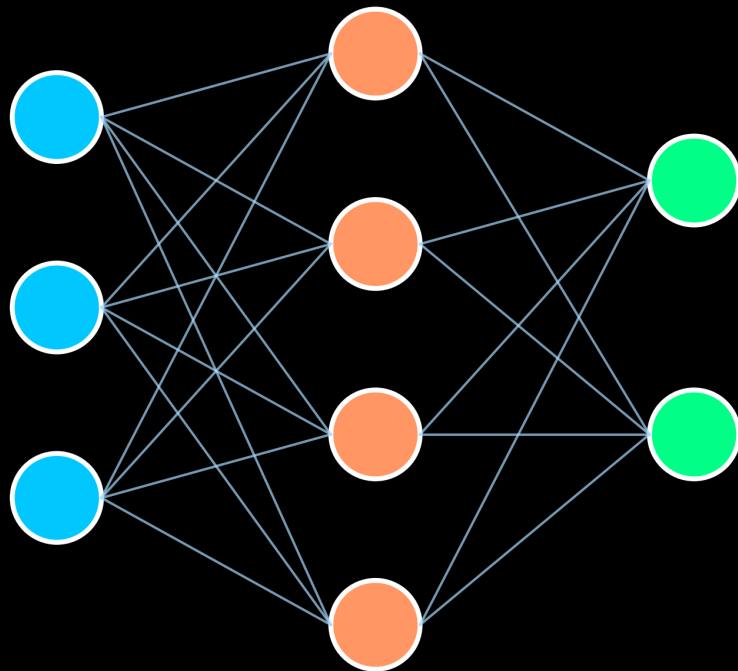
Contents of This Video

In this video, we will cover:

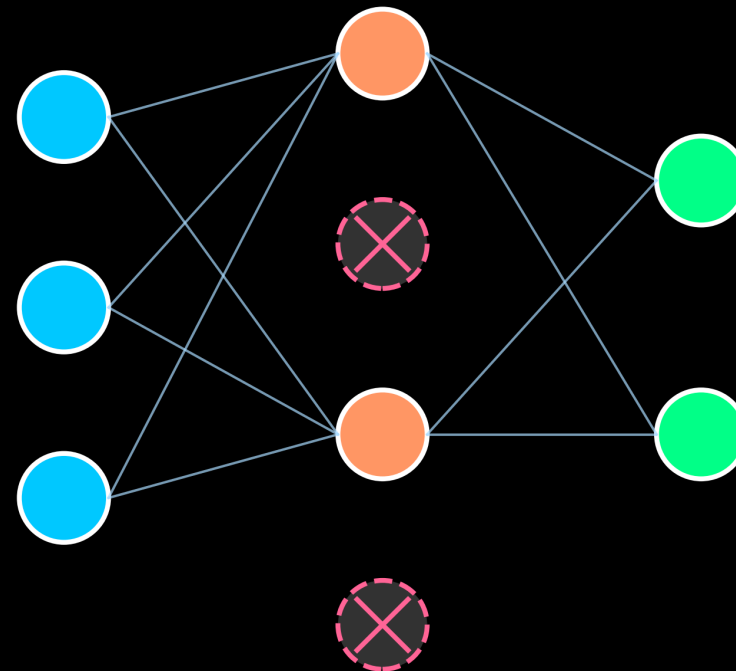
- What is dropout?
- How dropout works during training
- Training vs. inference behavior
- Why dropout prevents overfitting
- Practical usage guidelines

The Dropout Mechanism

Standard Network
(All neurons active)



With Dropout
(Some neurons dropped)



Dropout During Training

At each training step:

1. For each neuron, flip a coin (probability p)
2. If heads, set activation to 0 (“drop” the neuron)
3. Continue forward and backward passes normally

Mathematically:

$$\tilde{a}_j = a_j \cdot m_j \quad \text{where } m_j \sim \text{Bernoulli}(1 - p)$$

Common dropout rate: $p = 0.2$ to 0.5

Training vs. Inference

During training:

- Drop neurons randomly with probability p
- Network sees different “thinned” versions each batch

During inference (testing):

- Use ALL neurons (no dropout)
- Scale activations by $(1 - p)$ to compensate

Or use “inverted dropout” during training:

- Scale activations by $\frac{1}{1-p}$ during training
- No scaling needed at inference

Why Dropout Works

Theory 1: Prevents co-adaptation

- Neurons can't rely on specific other neurons
- Must learn independently useful features
- Forces redundant representations

Theory 2: Implicit ensemble

- Each dropout pattern = different “thinned” network
- Training sees exponentially many networks
- Inference averages their predictions

Dropout in Practice

Where to apply:

- Typically after fully connected layers
- Sometimes after convolutional layers (less common)
- Never on the output layer

Typical rates:

- FC layers: 0.5
- Convolutional layers: 0.2-0.3

What We've Covered

In this video, we've learned:

- Dropout randomly sets activations to zero during training
- Each batch sees a different “thinned” network
- Prevents neurons from co-adapting
- Acts like an implicit ensemble of many networks
- Use rate 0.2-0.5 on FC layers, none on output layer